Tobias Großbölting

Ph.D. Candidate in Economics, University of Mannheim

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Placement Director: Prof. Christoph Rothe c.rothe@uni-mannheim.de

RESEARCH FIELDS

Econometrics
Causal inference, regression discontinuity designs, nonparametric estimation, bias-aware inference

DOCTORAL STUDIES

University of Mannheim, CDSE *September 2021 – June 2027 (Expected)*
Ph.D. Candidate in Economics
Supervisors: Prof. Christoph Rothe and Prof. Paul Goldsmith-Pinkham

Yale University *2024 – 2025*
Visiting Ph.D. Student.

PRE-DOCTORAL STUDIES

Humboldt University of Berlin *2018 – 2021*
M.Sc. Economics.

University of Copenhagen *2019*
Visiting Student, Department of Economics.

Humboldt University of Berlin *2015 – 2018*
B.Sc. Economics.

University of Warwick *2017 – 2018*
Visiting Student, Department of Economics.

REFERENCES

Christoph Rothe
Professor of Economics
University of Mannheim
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Paul Goldsmith-Pinkham
Associate Professor of Finance
Yale School of Management
paul.goldsmith-pinkham@yale.edu

Soonwoo Kwon
Assistant Professor of Economics
Brown University
soonwoo_kwon@brown.edu

JOB MARKET PAPER

Composition Bias and Covariate Adjustment in Regression Discontinuity Designs (pdf)
In regression discontinuity designs covariates can be imbalanced in the samples used for estimation, even if their distribution is continuous in the running variable. I demonstrate that under such covariate imbalance the standard local polynomial RD estimator (base RD) implicitly compares units with different covariate values. This induces an additive term in the bias of the base RD estimator, which I refer to as composition bias. To reduce composition bias, I argue that researchers should adjust for pre-treatment covariates so that the estimator compares only units with similar covariate values. Specifically, I propose the Stratified RD estimator and prove that it has asymptotically weakly lower variance and worst-case bias than the base RD estimator, which leads to shorter bias-aware confidence intervals for the average treatment effect at the cutoff. In a simulation calibrated to replication data from a study of a merit-based financial aid program in Colombia, stratifying on parental education reduces the bias relative to the base RD estimator by 14–23%, and stratifying on both parental education and whether a student lives in a metropolitan area reduces it by 25–30%, depending on the bandwidth.

WORK IN PROGRESS

Segment Fixed Effects in Spatial Regression Discontinuity Designs
joint with Soonwoo Kwon, Michael Pollmann and Aurel Rochell

Making a Difference:
Unobservables and Pre-Treatment Outcomes in Regression Discontinuity Designs

Generic Machine Learning Inference on Heterogeneous Treatment Effects in RD Designs
joint with Paul Goldsmith-Pinkham

Narratives: The Effect of Political Advertising on Attitudes towards Migration
joint with Johannes Matzat

TEACHING EXPERIENCE

University of Mannheim

Econometrics I (PhD-level), Prof. Christoph Rothe	*Fall 2025*
Introduction to Econometrics, Prof. Carsten Trenkler	*Spring 2023, 2024*
Statistics II, Dr. Toni Stocker	*Fall 2022, 2026*

Humboldt University of Berlin

Macroeconomics I, Prof. Lutz Weinke	*Fall 2020*
Macroeconomics II, Prof. Michael Burda	*Spring 2020*
Macroeconomics I, Prof. Michael Burda	*Fall 2019*

CONFERENCE AND SEMINAR PRESENTATIONS

2026
NASM of the Econometric Society, Emory University; ENTER Seminar, UCL; ENTER Jamboree, UC3M; HKMetrics Workshop, Karlsruhe Institute of Technology

2025
Econometrics Lunch, Brown University; Econometrics Prospectus Lunch, Yale University; CIML Training Conference: Frontiers of Causal Inference and Machine Learning, IMT Lucca

2024
Econometrics Prospectus Lunch, Yale University; Bonn-Mannheim-Munich Workshop in Microeconometrics, University of Mannheim; Workshop on Networks and Political Economy, Paris 1 Panthéon-Sorbonne

2022
FHM+ Workshop on Development Economics, Heidelberg University

SUMMER SCHOOLS

Gerzensee Summer School: ML for Treatment Effects & Structural Equation Models *2024*
Prof. Victor Chernozhukov

Northwestern Advanced Causal Inference Workshop *2024*
Prof. Peter Hull, Prof. Christian Hansen, Prof. Andrew Goodman-Bacon

CEMFI Summer School: Causal Inference for Medical and Social Science *2023*
Prof. Miguel Hernán

OTHER PROFESSIONAL EXPERIENCE

Research Assistance

Prof. Christoph Rothe, University of Mannheim	*2023 – 2024*
Prof. Michael Burda, Humboldt University of Berlin	*2020 – 2021*
Prof. Alexander Stomper, Humboldt University of Berlin	*2018 – 2019*

AWARDS AND DISTINCTIONS

DAAD Forschungsstipendium für Doktorandinnen und Doktoranden *2024 – 2025*
Scholarship awarded by the German Academic Exchange Service (DAAD) for my research visit to Yale University.

Studienförderung der Friedrich-Ebert-Stiftung *2017 – 2021*
Government-funded scholarship granted for academic excellence and social engagement.

Portfolio Management Program – P & K Pühringer Foundation, ESMT and HU Berlin *2018 – 2020*
Group-based active portfolio management; certificate and prize for excellent performance.
Mentor: Dr. Gerald Leitner.

MISCELLANEOUS

Programming	R, Python, Java
Software	LaTeX, Zotero
Languages	German (native), English (fluent), Spanish and Italian (beginner)

Last updated: November 2026